Zomato_Data_Analysis_Recommendation_System

Python-Based Restaurant Recommendation Using EDA and Machine Learning
Python | Data Analysis | Machine Learning | K-Means Clustering

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Project Overview

- Analyze Zomato dataset using Python
- Perform data cleaning and preprocessing
- Conduct Exploratory Data Analysis (EDA)
- Extract meaningful insights from data
- Visualize patterns using graphs and charts
- Build a basic recommendation system

Objectives

- Understand real-world dataset
- Perform Exploratory Data Analysis (EDA)
- Visualize trends and patterns
- Apply basic Machine Learning concepts

Dataset Description

Dataset: Zomato Restaurants
Source: Kaggle
City Covered: Bangalore
Total Records: 51,717
Total Features: 17

Link:https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants

Feature Categories:

- Restaurant Information :
  Name, Location, Address
- Food & Services :
  Cuisines, Restaurant Type
  Online Order, Table Booking
- User Feedback :
  Rating, Votes, Reviews
- Cost Details :
  Approx Cost for Two
- Additional Features :
  Listed in (Type & City)
  Dish Liked

missing values in dataset

```
url                               0
address                           0
name                              0
online_order                      0
book_table                        0
rate                           7775
votes                             0
phone                          1208
location                         21
rest_type                       227
dish_liked                    28078
cuisines                         45
approx_cost(for two people)     346
reviews_list                      0
menu_item                         0
listed_in(type)                   0
listed_in(city)                   0
dtype: int64
```

Handling Missing Values

- To improve data quality and ensure accurate recommendations
- Handle missing values based on feature importance

Approach

- Prioritized columns based on their impact on the recommendation system
- Applied:
  - Imputation using derived data (e.g., ratings from reviews)
  - Statistical imputation (mean/median/mode) for missing values
  - Column removal for irrelevant or highly incomplete features
  - Minimal row removal where data was unavailable

Handling Missing Values – Rate

- rate is a key feature for recommendation
- Directly impacts restaurant ranking and user suggestions

Method

- Extracted ratings from reviews_list using RegEx (e.g., “Rated 4.0”)
- Converted extracted values into numeric format and computed average rating
- Filled missing rate values using this average
- Removed rows where rating was still unavailable

```
ratings = re.findall(r'Rated (\d\.\d)', str(review))
avg = sum(ratings)/len(ratings)
```

	reviews_list	extracted_rating
84	[('Rated 4.0', 'RATED\n Good Location. Small ...	4.0
90	[('Rated 1.0', 'RATED\n Do not order anything...	1.0
91	[]	NaN
92	[]	NaN
107	[]	NaN

Handling Missing Values – Cuisines

- cuisines is essential for food preference-based recommendations

Method

- Extracted only the primary cuisine (first value from list)
- Filled missing values using the most frequent cuisine (mode)

```
d2['cuisines'] = d2['cuisines'].str.split(',').str[0]
d2['cuisines'] = d2['cuisines'].fillna(d2['cuisines'].mode()[0])
```

	Before	After
0	North Indian, Mughlai, Chinese	North Indian
1	Chinese, North Indian, Thai	Chinese
2	Cafe, Mexican, Italian	Cafe
3	South Indian, North Indian	South Indian
4	North Indian, Rajasthani	North Indian

Handling Missing Values – Approx Cost(for two people)

- Used for budget-based recommendations
- Helps users filter restaurants based on affordability
- Improves personalization by matching user spending preferences

Method

- Removed commas to standardize the data
- Converted values into numeric format
- Replaced missing values using the median cost

Why Median?

- Less affected by outliers compared to mean
- Provides a more robust estimate for cost data

Median Values Used: 450.0

Handling Missing Values – rest_type

- Helps categorize restaurants (e.g., Casual Dining, Cafe)
- Useful for filter-based recommendations

Method

- Filled missing values using the most frequent category (mode)

Why Mode?

- **Suitable for categorical data**
- **Preserves the most common restaurant type**

Mode Value Used: Quick Bites

Feature Selection & Removal

Removed Features

- phone → Not useful for analysis
- menu_item → Mostly missing values
- url → Irrelevant for modeling
- address → Not required for recommendations
- dish_liked → High missing values (~22,000)

Selected Features

- Numerical: votes, cost, rate
- Categorical: location, cuisines, restaurant type
- Binary: online_order, book_table

```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
location                       0
rest_type                      0
cuisines                       0
approx_cost(for two people)    0
reviews_list                   0
listed_in(type)                0
listed_in(city)                0
dtype: int64
```

Skewness & Kurtosis Analysis

Feature	Skewness	Kurtosis
Votes	7.19	80.77
Cost	2.51	9.75
Rate	-0.33	-0.01

Skewness (Measure of Asymmetry)

Skewness tells us whether the data is symmetrical or tilted (skewed) to one side.

- Positive skew (+) → Tail on the right (few large values)
- Negative skew (–) → Tail on the left (few small values)
- 0 skew → Perfectly symmetrical

Our Results :

- Votes → 7.1966 (Highly Positive Skew)
- Approx Cost → 2.5152 (Moderate Positive Skew)
- Rate → -0.3286 (Slight Negative Skew)

Interpretation:

Votes (7.19)

- Extremely right-skewed
- Most restaurants have low votes, but few have very high votes (outliers)
- Matches our histogram & boxplot (many extreme points)

Approx Cost (2.51)

- Right-skewed
- Most restaurants are affordable, but some are very expensive

Rate (-0.32)

- Slight left skew
- Ratings are mostly high (3–4.5) with few low ratings

Skewness Analysis

Kurtosis (Measure of Tailedness / Outliers)

- Kurtosis indicates how heavy the tails are → presence of outliers
- In pandas (excess kurtosis):
  - > 0 → High (Leptokurtic) → Many outliers, sharp peak
  - ≈ 0 → Normal distribution → Moderate distribution
  - < 0 → Low (Platykurtic) → Flat distribution, fewer outliers

Our Results:

- Votes → 80.77 (Extremely High Kurtosis)
- Approx Cost → 9.76 (High Kurtosis)
- Rate → -0.007 (Near Normal)

Kurtosis Analysis

Interpretation:

Votes (80.77)

- Extremely high kurtosis
- Indicates huge number of extreme outliers
- Some restaurants are very popular compared to most

Approx Cost (9.76)

- High kurtosis
- Presence of expensive outliers
- Most values are clustered, but few are very high

Rate (-0.007)

- Almost normal distribution
- Ratings are well-balanced and consistent
- Very few extreme values

Outlier Detection & Treatment

- Outliers detected using IQR (Interquartile Range) method :

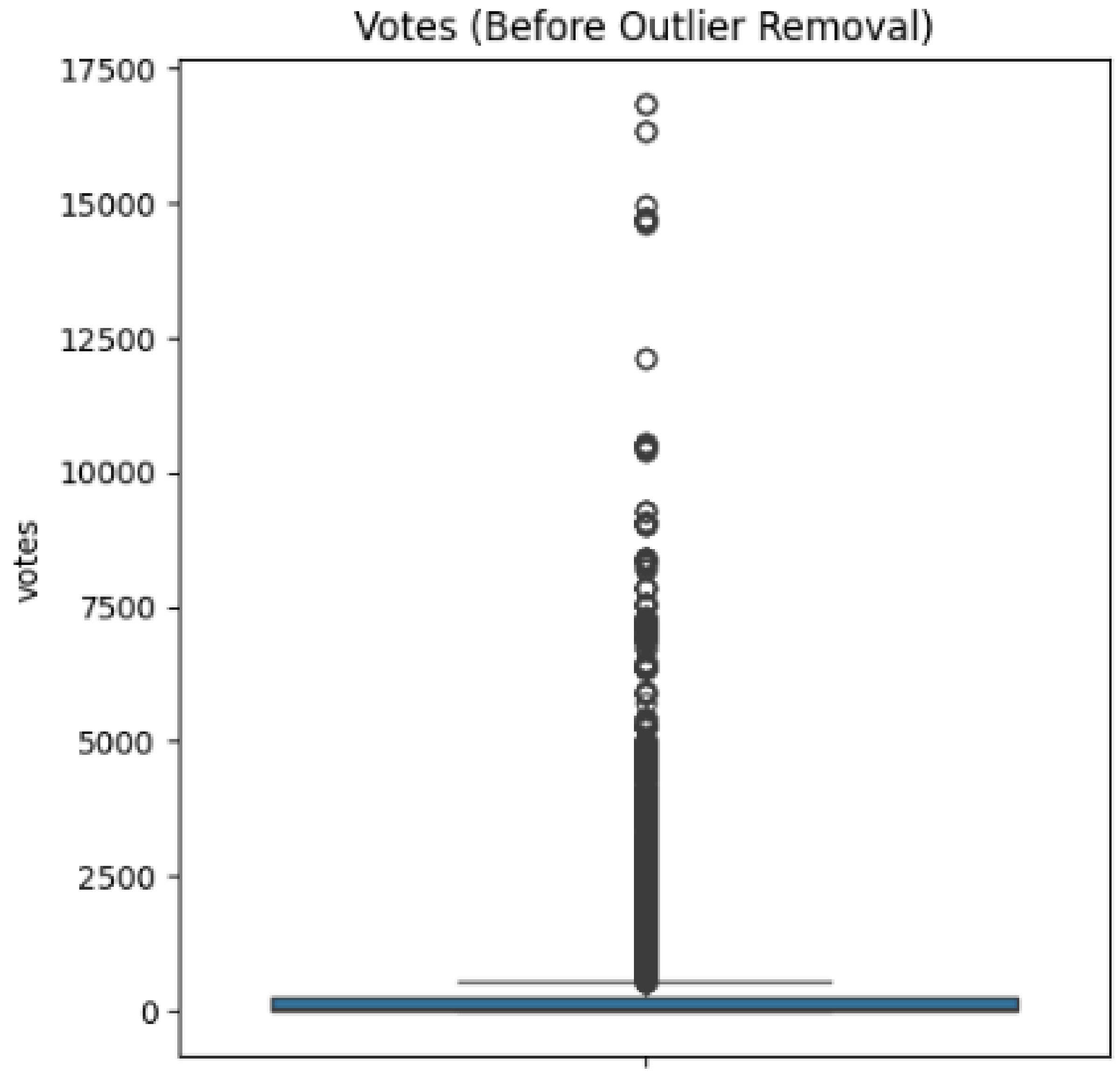
  Applied on: votes, cost, and rate features
- 5% threshold used:

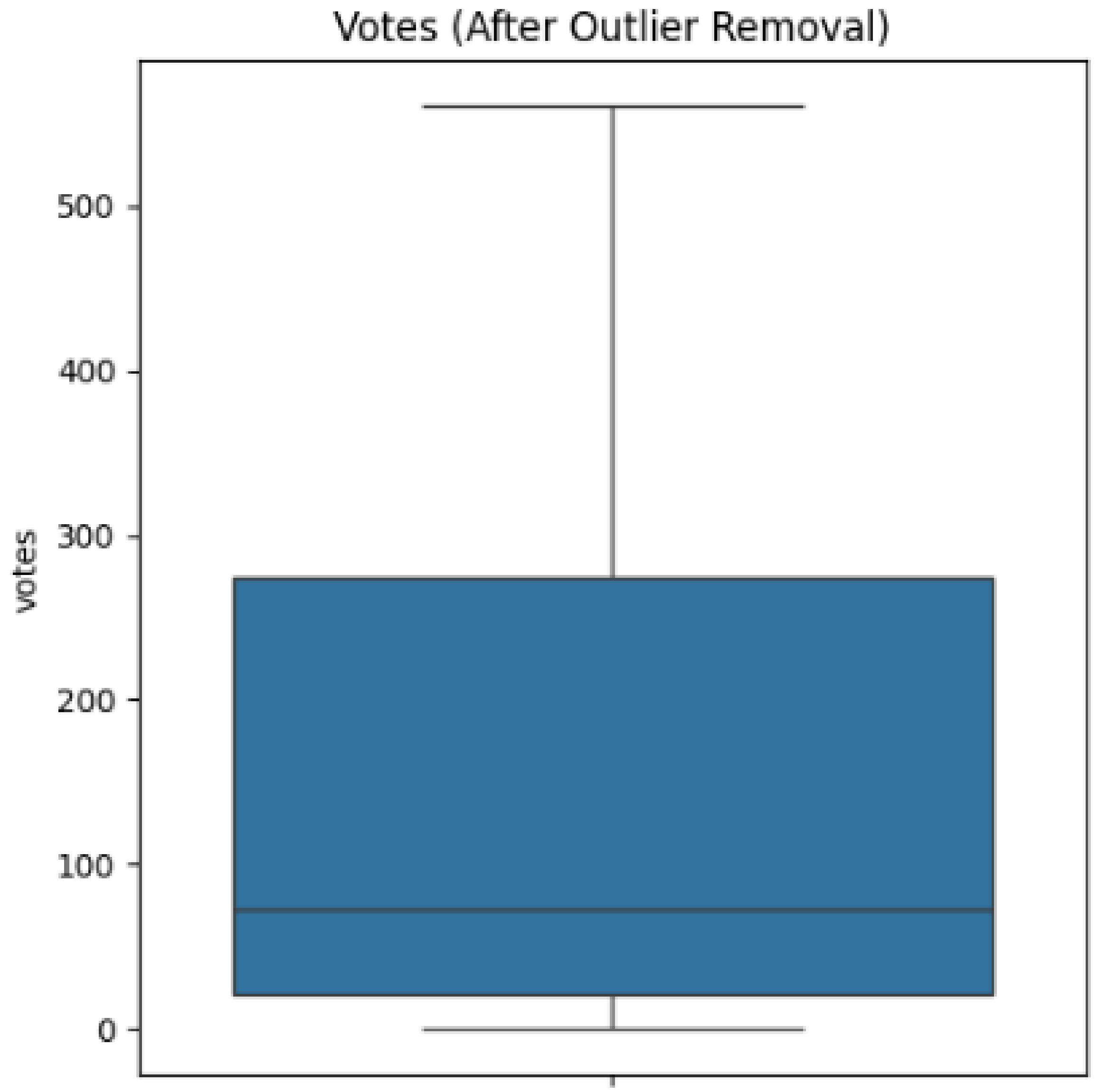
  < 5% → Outliers removed

  ≥ 5% → Outliers capped
- Boxplots were used to visualize data distribution before and after cleaning.
- Feature-wise treatment applied to preserve important data
- Improved data consistency for better model performance

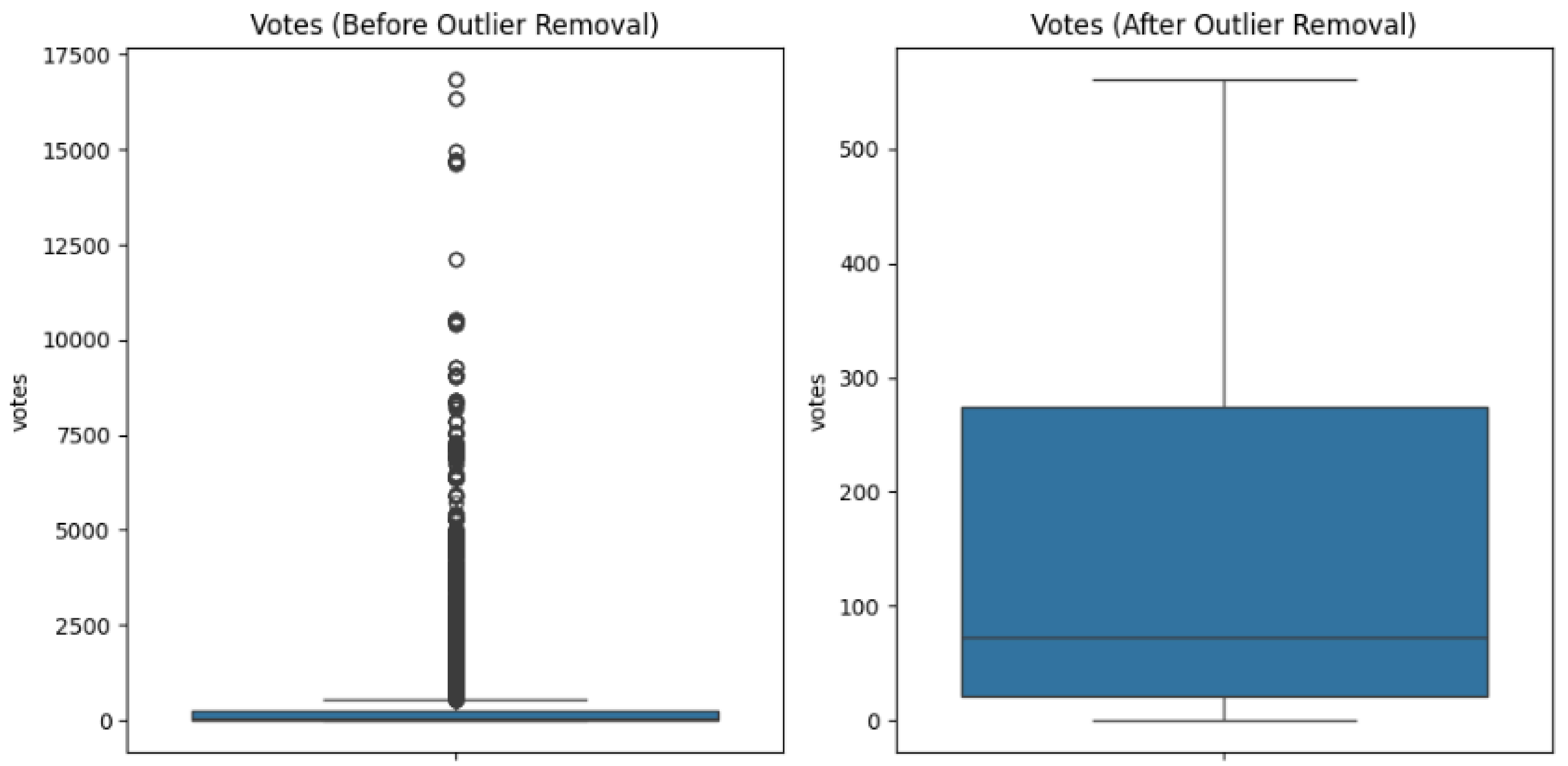
```
votes: 6164 outliers (13.27%)
→ Capping outliers

approx_cost(for two people): 3228 outliers (6.95%)
→ Capping outliers

rate: 187 outliers (0.40%)
→ Removing outliers
```

Outlier Removal – Votes

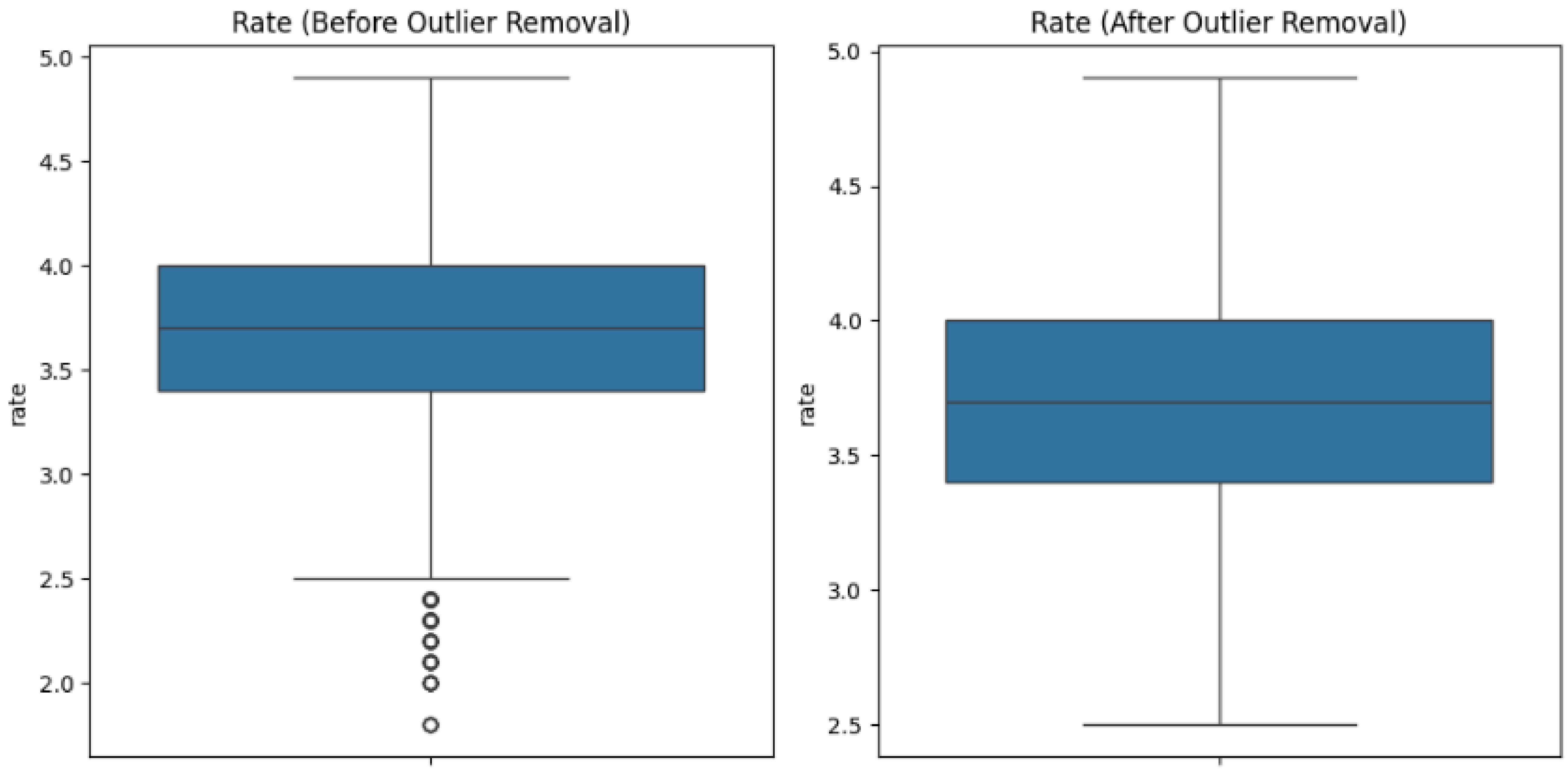




Outlier Removal – Approx Cost (for two people)

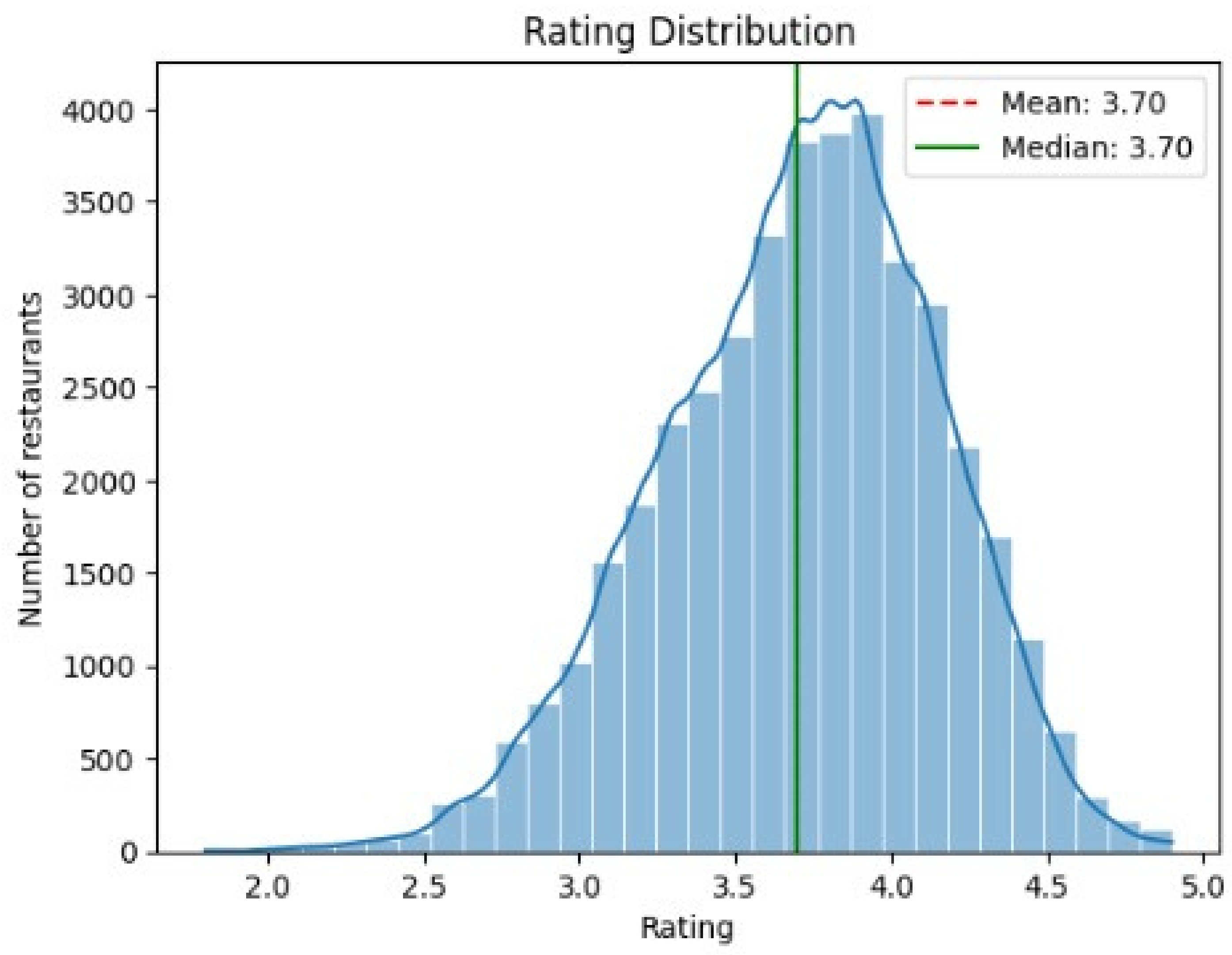


Outlier Analysis – Rating



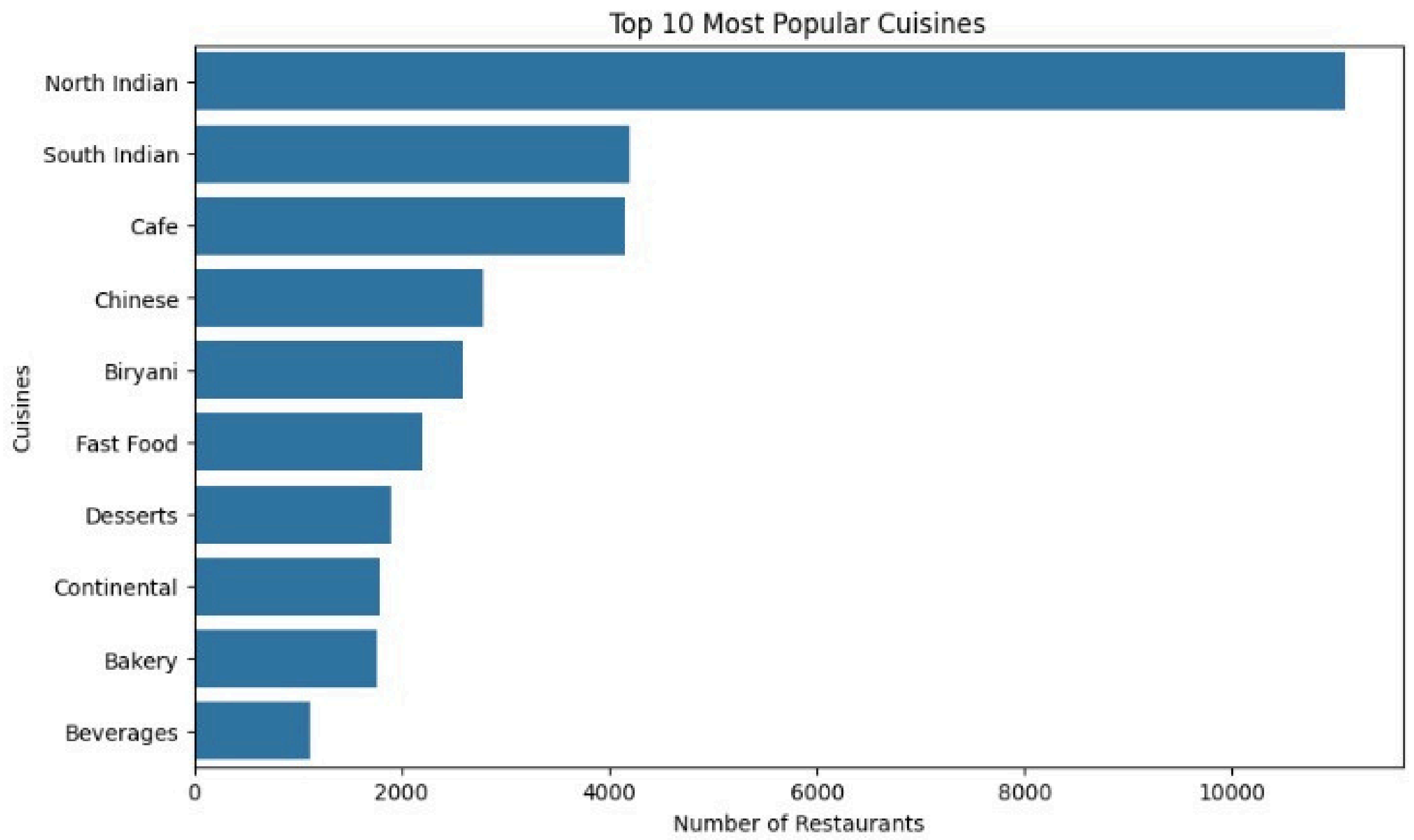
Data Visulization Techniques Used

Rating Distribution



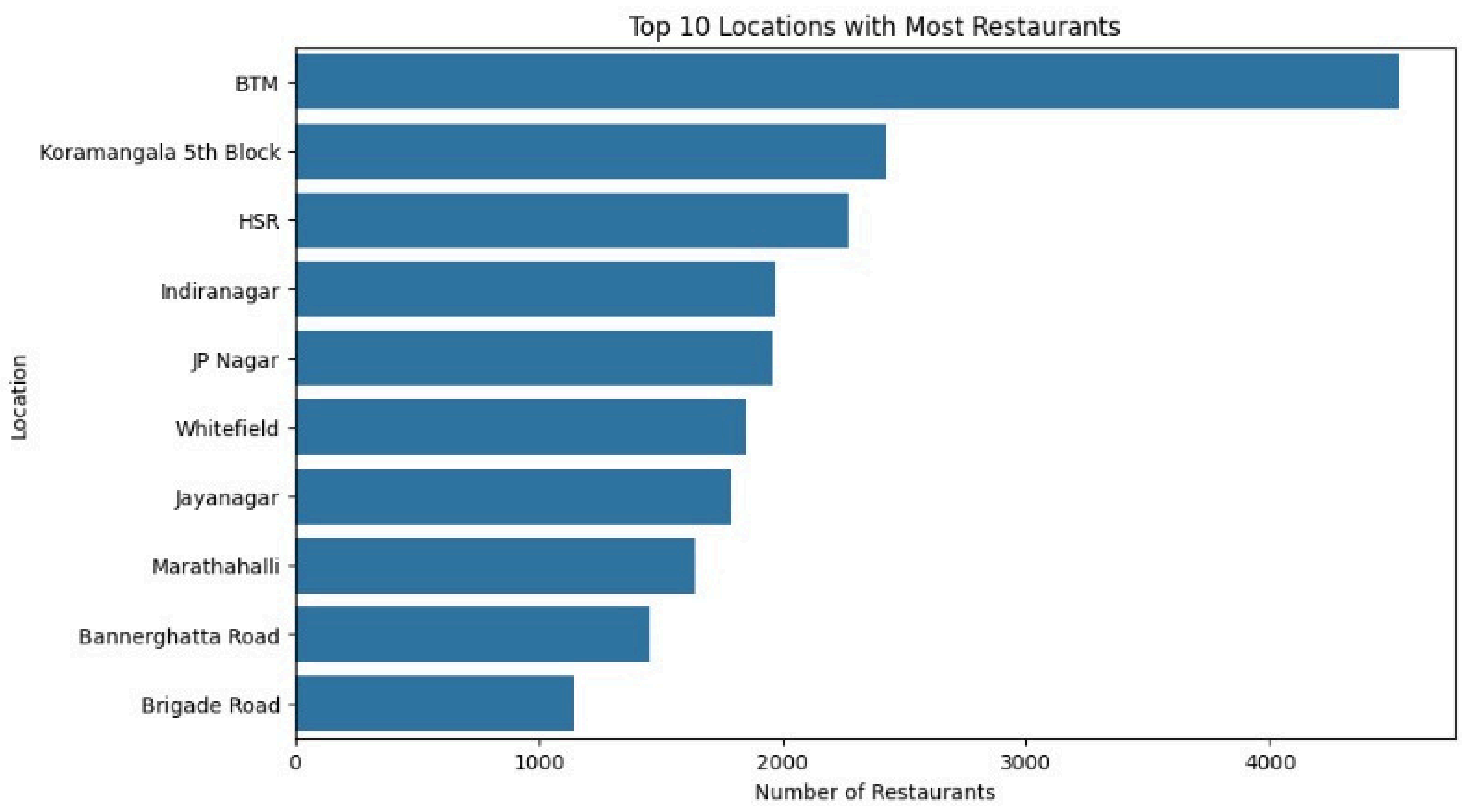
- Ratings follow a near normal distribution
- Mean ≈ Median (~3.7)
- Most restaurants rated between 3.5 – 4.2

Top 10 Cuisines



- North Indian cuisine is most popular
- Followed by South Indian and Cafe
- Useful for cuisine-based recommendations

Restaurant Distribution by Location

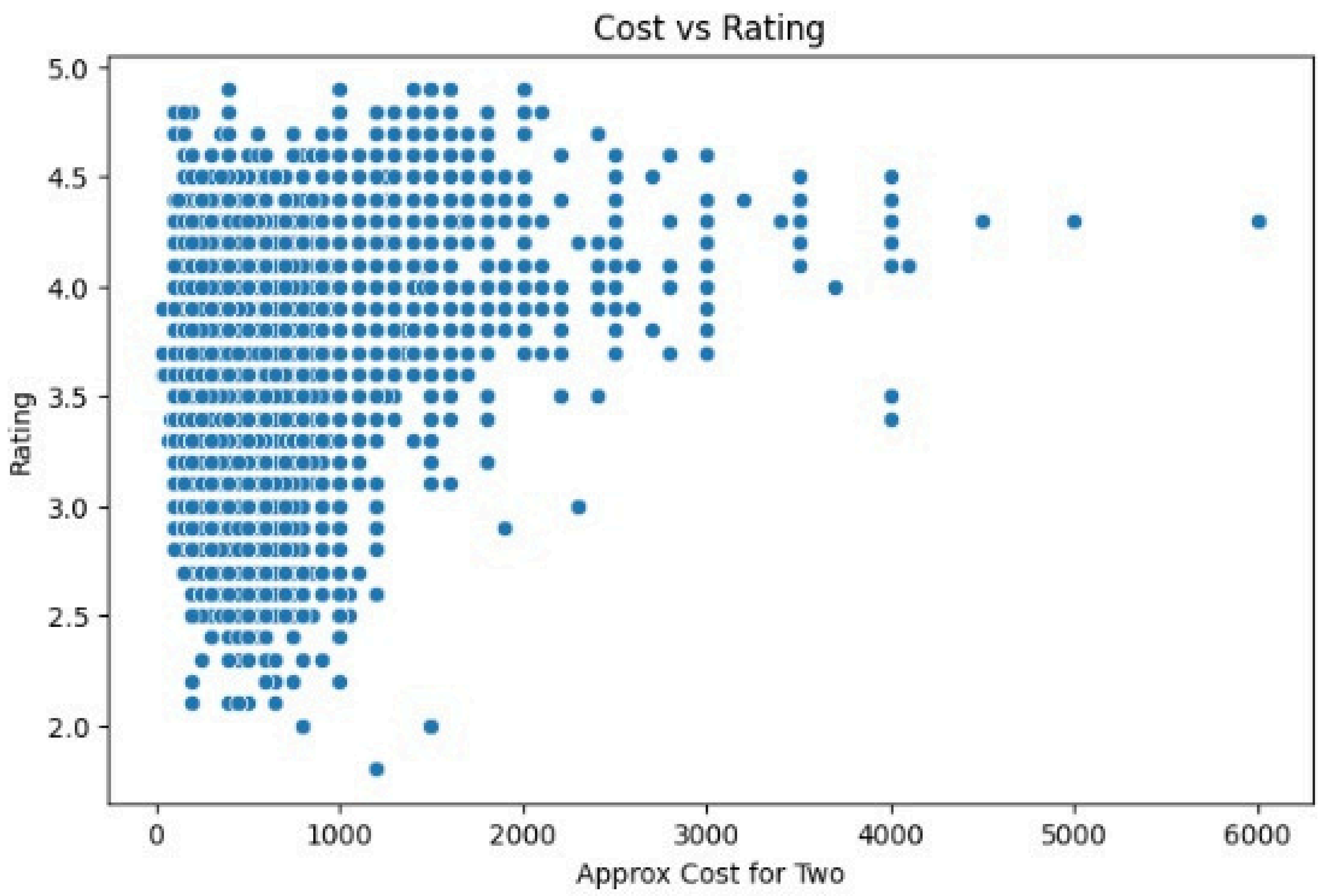


- BTM has highest number of restaurants
- Followed by Koramangala & HSR
- Indicates high-demand food areas

Impact of Online Ordering on Ratings

- Restaurants with online ordering have slightly higher ratings
- Overall rating distribution is similar
- Online availability may improve user experience

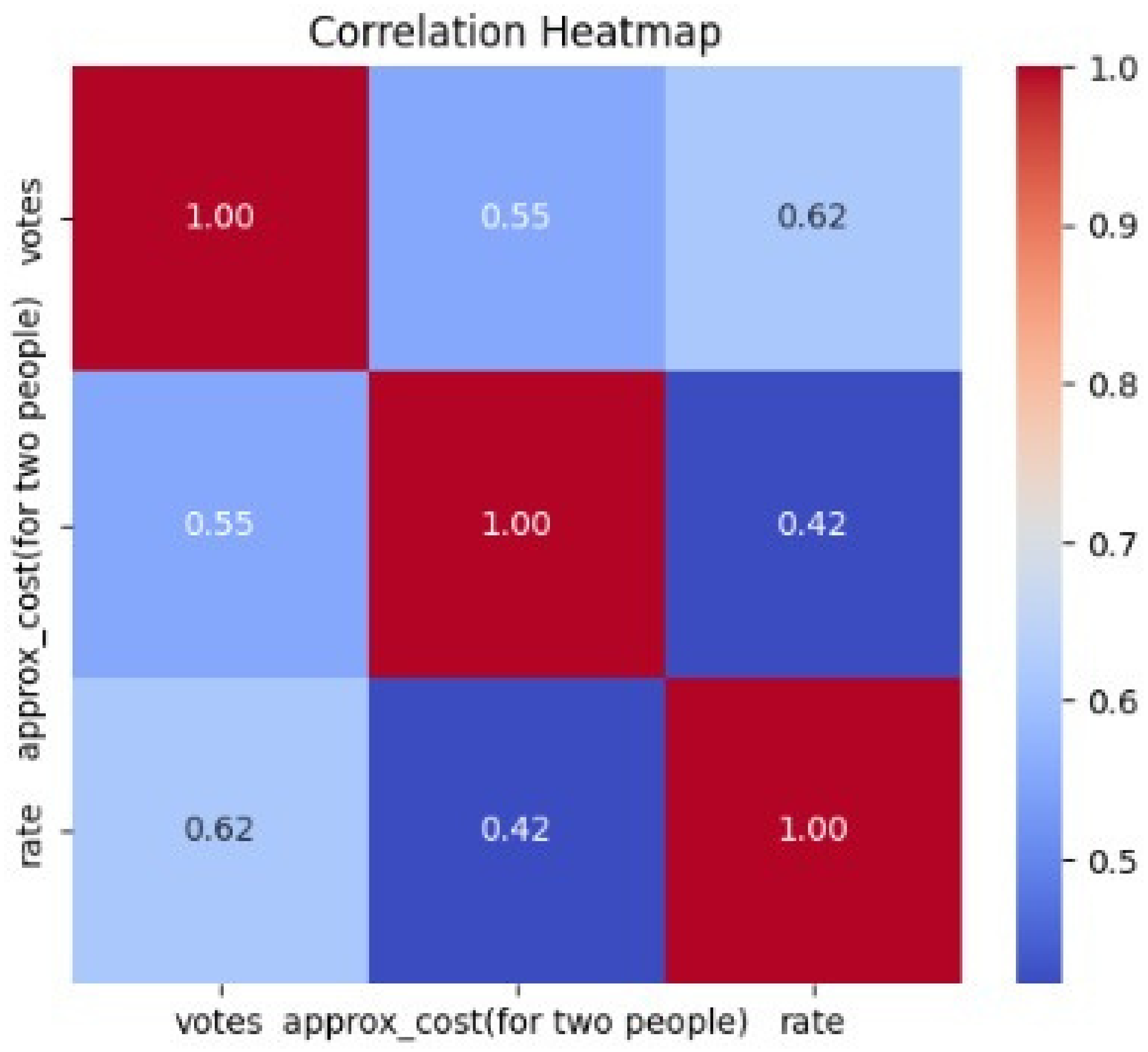
Cost vs Rating Analysis



- No strong correlation between cost and rating
- Expensive restaurants are not always highly rated
- Ratings depend on multiple factors beyond price

Feature Correlation Analysis

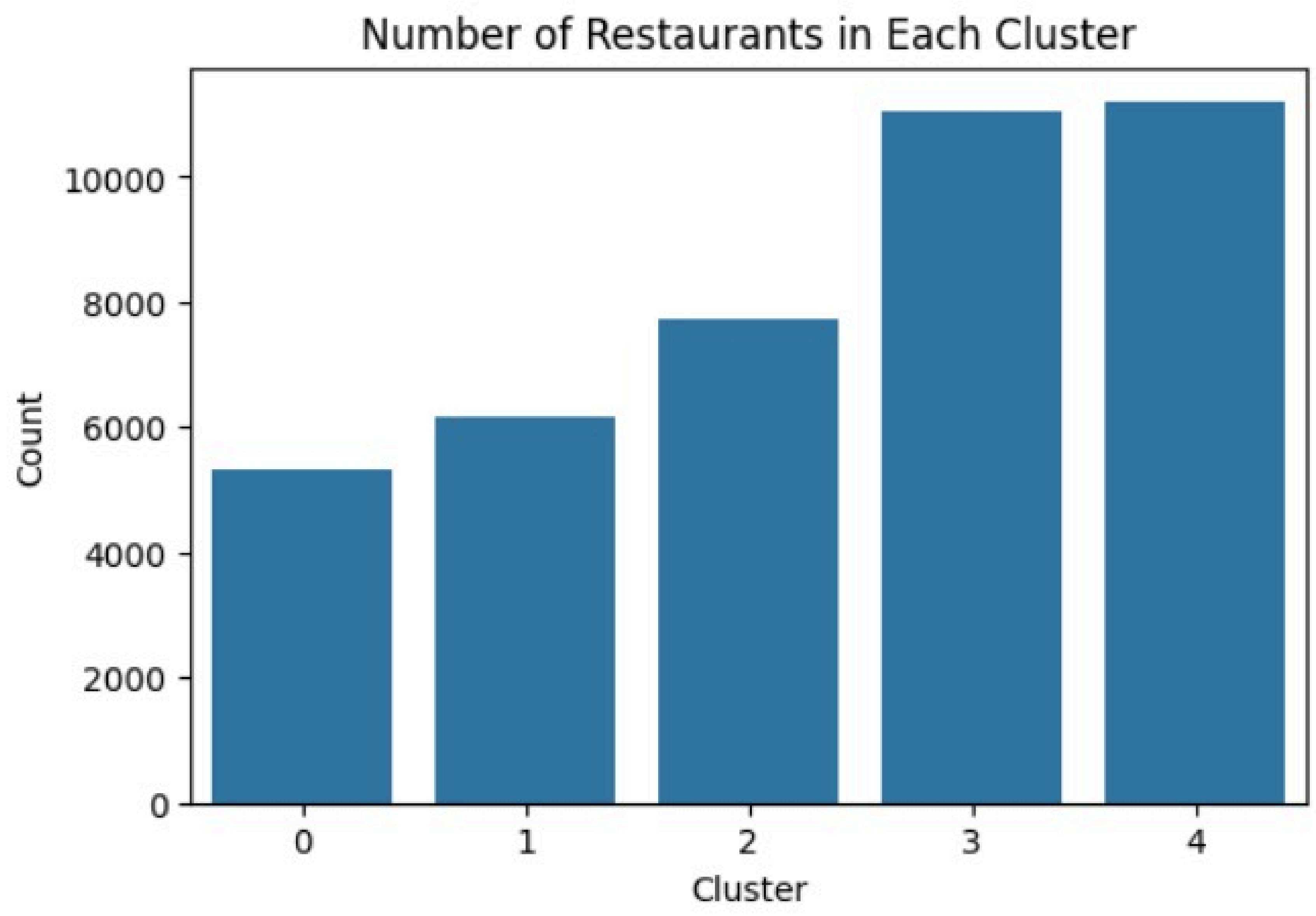
- Correlation computed after outlier treatment
- Weak relationship between cost and rating
- Slight positive correlation between votes and rating
- No strong linear relationships observed



KMeans Clustering – Methodology & Results

KMeans Clustering – Methodology

- Applied K-Means clustering (K = 5) to group similar restaurants
- Features used for clustering:
- Rating (quality)
- Votes (popularity)
- Cost (price level)
- Location, cuisine, type (categorical)
- Online order & table booking (services)
- All features were encoded & scaled for fair contribution
- Elbow Method used to select optimal clusters (K = 5)



Cluster Interpretation

cluster	rate	votes	approx_cost(for two people)
0	4.113030	462.532433	695.494938
1	4.158986	427.435476	1087.945272
2	3.433372	44.957768	399.270309
3	3.605462	78.176511	540.259987
4	3.553120	81.415788	339.880744

- Cluster 0 → Popular mid-range restaurants
- Cluster 1 → Premium high-end restaurants
- Cluster 2 → Low popularity restaurants
- Cluster 3 → Average performing restaurants
- Cluster 4 → Budget restaurants

Restaurant Recommendation System

User provides input:
Cuisine, Location, Budget

Step 1: Filtering :
Restaurants filtered based on user preferences

Step 2: Clustering (KMeans) :
Similar restaurants identified using cluster

Step 3: Ranking :
Restaurants ranked by rating and votes

Step 4: Final Output :
Top 3–5 unique restaurants recommended

Recommendation Example :

Input:

Cuisine: South Indian

Location: BTM

Budget: ₹800

Output:

Sri Vishnu Park

```
Enter cuisine:  South Indian
Enter location:  BTM
Enter max budget:  800
                              name location      cuisines  rate   votes  \
37173              Sri Vishnu Park      BTM  South Indian   3.8   102.0
9123                   Swaadam Veg      BTM  South Indian   3.6    34.0
879     Sankranthi Veg Restaurant      BTM  South Indian   3.0   102.0

       approx_cost(for two people)
37173                        350.0
9123                         300.0
879                          600.0
```

Key Insights from Analysis :

- Most restaurants have ratings between 3.5 – 4.2, indicating moderate customer satisfaction
- North Indian cuisine is the most popular category
- Areas like BTM, Koramangala, and HSR have high restaurant density
- Cost and rating have weak correlation, showing price does not guarantee quality
- Restaurants with higher votes tend to have better ratings, indicating popularity impact

Future Scope:

- Incorporate user reviews and sentiment analysis for better recommendations
- Develop a web or mobile interface for real-time user interaction
- Use advanced recommendation techniques like collaborative filtering
- Add real-time data updates for dynamic recommendations
- Improve model using user feedback and personalization

Conclusion :

- Successfully analyzed restaurant dataset and identified key patterns
- Applied KMeans clustering to group similar restaurants
- Built a recommendation system based on user preferences and clustering
- Provided meaningful and personalized restaurant suggestions
- The system can be extended for real-world applications